Antigen-Specific Antibody Design and Optimization with Diffusion-Based Generative Models for Protein Structures

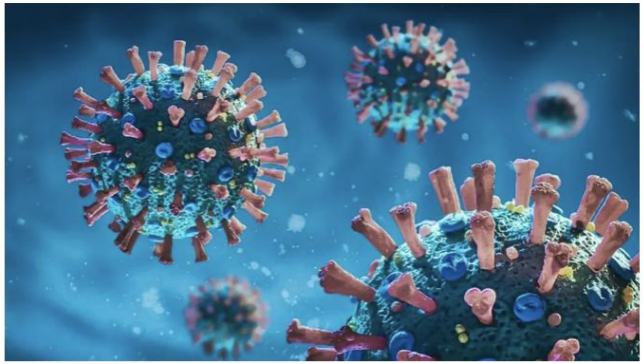
Shitong Luo, Yufeng Su, Xingang Peng, et al.

Presenter: Christine Wu; Date: 12/2/2024

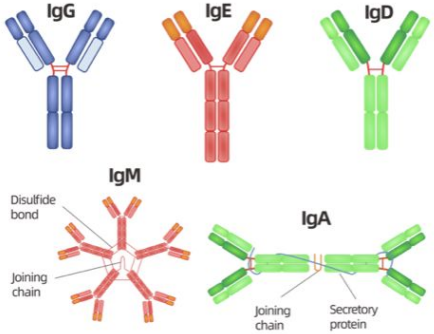
Introduction

Introduction

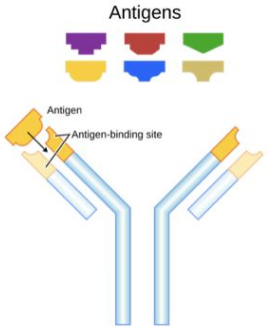
What are antibodies?



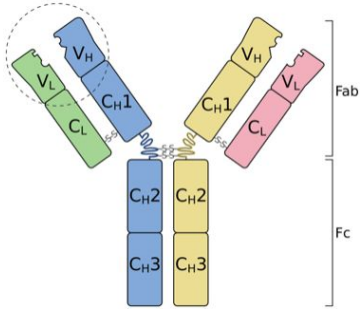
Antibodies against the SARS-CoV-2 virus.



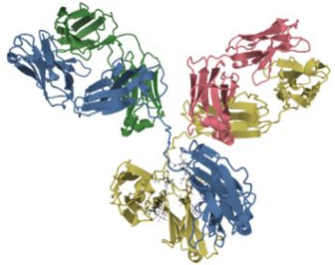
Different Types of Antibodies.



Each antibody binds to a specific antigen in a highly specific interaction analogous to a lock and key.



Schematic structure of an antibody: two heavy chains (blue, yellow) and the two light chains (green, pink). The antigen binding site is circled.



A more accurate depiction of an antibody (3D structure at RCSB PDB)

Antibody Structure, classes and functions_AntibodySystem. (n.d.). https://www.antibodysystem.com/archive/48.html

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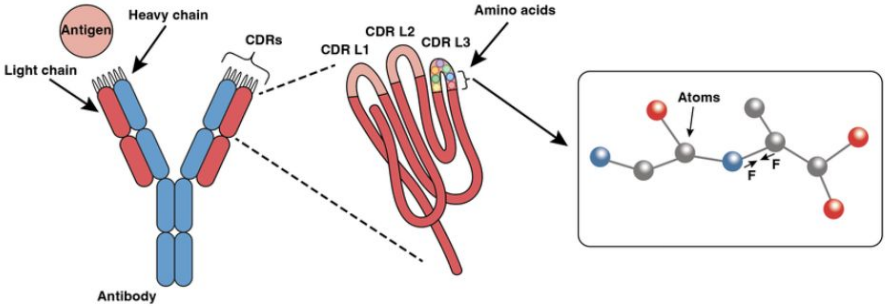
Introduction

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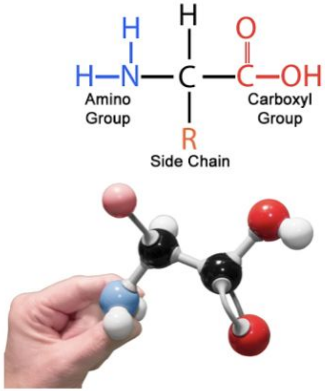
Kulytė, P., Vargas, F., Mathis, S. V., Wang, Y. G., Hernández-Lobato, J. M., & Liò, P. (n.d.). Improving Antibody Design with Force-Guided Sampling in Diffusion Models. *alphaXiv*. https://www.alphaxiv.org/abs/2406.05832v1

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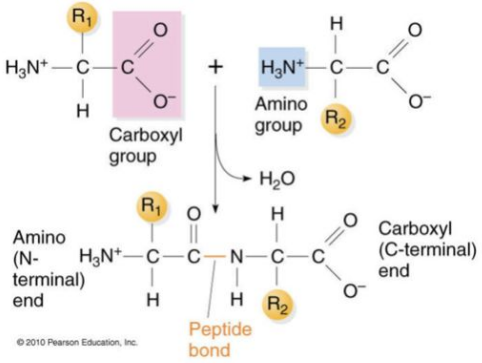
Importance of Complementarity Determining Regions (CDRs)



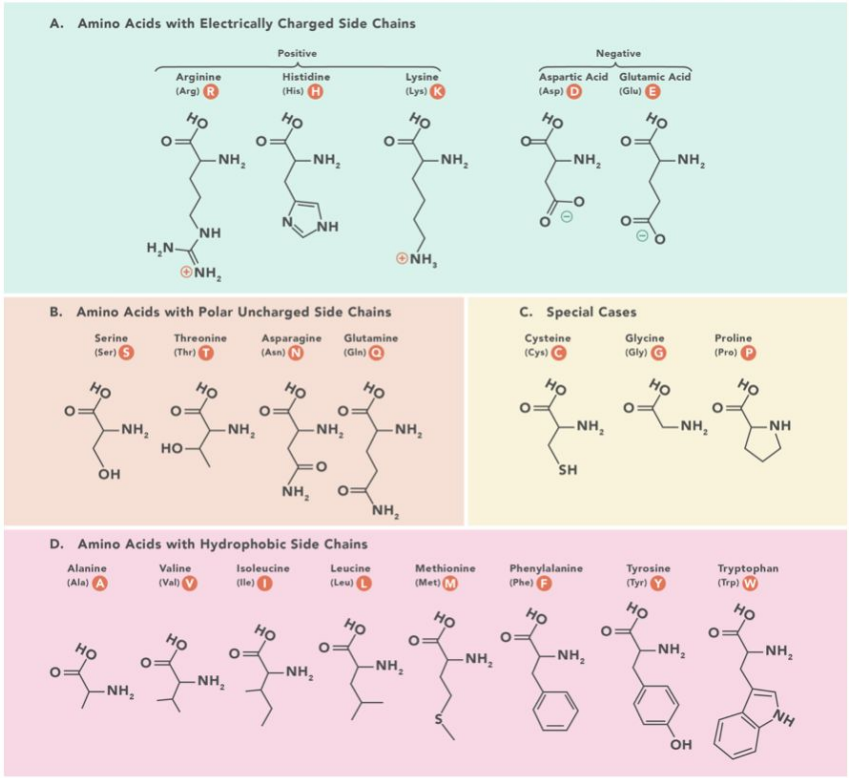
The antigen-binding region comprises six complementarity-determining regions (CDRs).



An amino acid.



A peptide formation through condensation rxn.



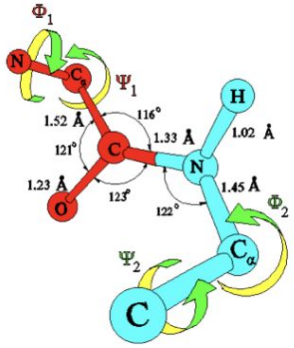
Amino Acid chart.

CHEM 440 - Dihedral angles. (n.d.). http://guweb2.gonzaga.edu/faculty/cronk/chem440pub/dihedral.html

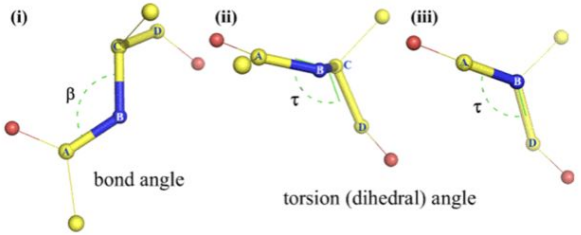
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Introduction

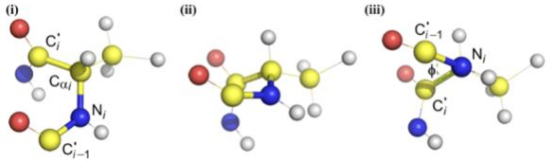
Importance of AA Orientations & Docking Effects



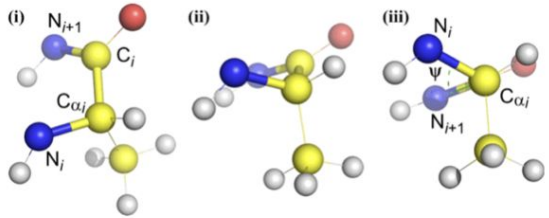
Polypeptide main chain dihedral angles: Phi (ϕ), Psi (ψ), and Omega (ω)



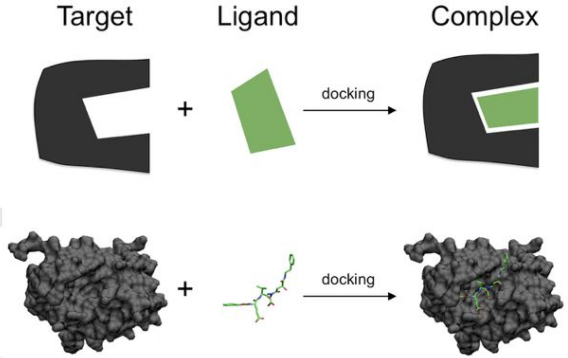
Phi (ϕ) - central bond between N(*i*) [the amide nitrogen of residue *i*] and C(alpha, *i*) [the alpha carbon of residue *i*]

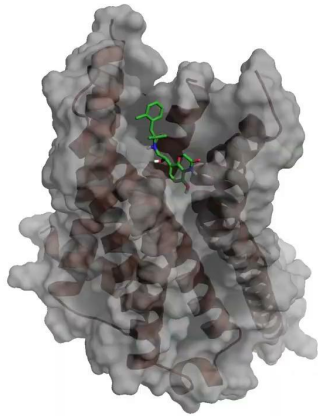


Psi (ψ) - central bond between C(alpha, i) and C(i) [the acyl carbon of residue *i*].



Omega (ω) - the peptide bond between the acyl carbon C(*i*) and N(*i*+1) is the central bond.





- Hydrogen Boding
- Electrostatic interactions
- Van der Waals forces
- Hydrophobic interactions

Introduction

Challenges in designing antigen-specific antibodies

Large Search Space
- Complementarity Determining Regions (CDRs) involve complex sequences and structures.
- A single CDR sequence with length L can have up to 20^L possible configurations, making exhaustive exploration infeasible.

3D Structure Dependency
- Effective antibody design must consider the 3D structure of both antigens and antibodies.
- Generating CDRs that precisely fit antigen structures in 3D space is a complex task.

Interaction Specificity
- Binding between antibodies and antigens is determined by specific side-chain interactions.
- Modeling the positions and orientations of these side chains accurately is critical.

Generalization to New Antigens
- Traditional models struggle to design antibodies for novel antigens due to overfitting to known structures.
- The design process must adapt to antigens with unknown or variable 3D conformations.

Optimization vs. De Novo Design
- In practical applications, optimizing existing antibodies for higher binding affinity is often preferred over designing entirely new ones.
- Balancing improvements in affinity while maintaining structural similarity is challenging.

Computational Inefficiency
- Many traditional methods rely on sampling-based approaches using energy functions.
- These approaches are computationally expensive and prone to getting trapped in local minima.

- **Key Takeaway of Challenges**
  - **Complexity of CDRs regions with different AA types**
  - **The binding interactions between antibodies & antigens (orientations & docking effects)**
  - **What is the best algorithm?**
    - **Computational efficiency**
    - **Computational expenses**

A deep generative model using diffusion-based methods



Objective

Objective

To develop **a deep generative model using diffusion-based methods** to **design and optimize** antibody Complementarity Determining Regions **(CDRs)** that:

1. Explicitly target specific antigen structures in 3D space.
2. Address key challenges in antibody design, such as:
   - Incorporating antigen **3D structures.**
   - Modeling amino acid **positions and orientations** for **precise interaction**.
   - **Optimizing** existing antibodies for improved performance.

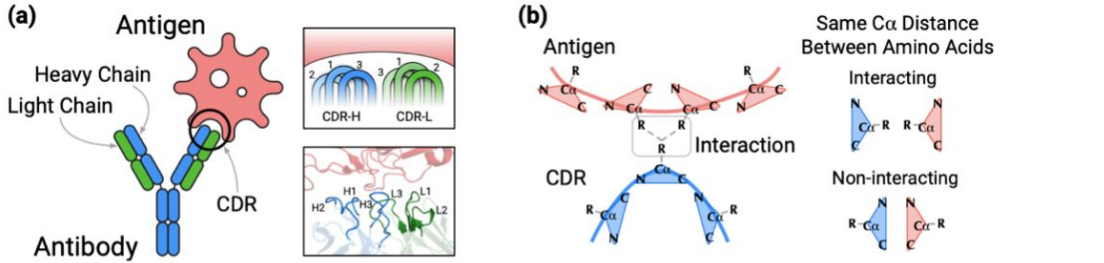


Figure 1: **(a)** Antibody-antigen complex structure and CDR structure. **(b)** The orientations of amino acids (represented by triangles) determine their side-chain orientations, which are key to inter-amino-acid interactions.

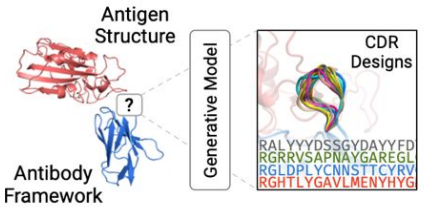


Figure 2: The task in this work is to design CDRs for a given antigen structure and an antibody framework.

Methods

Methods

I: Representation of Amino Acids and Problem Formulation

An amnio acid in a protein complex is represented by:

- Its **type**: $s_i \in \{A, C, D, E, F, G, H, I, K, L, M, N, P, Q, R, S, T, V, W, Y\}$
- The coordinate of its **α-carbon atom:** $x_i \in \mathbb{R}^3$
- Its **orientation:** $O_i \in SO(3)$

  * Here, i = 1, …, N, where N is the total number of amino acids in the protein complex.

Focus of the study:

- Give the antigen structure and antibody framework, the goal is to design the CDRs of the antibody.
- The **CDR region (R)** is represented as a set of **amnio acid types, coordinates, and orientations.**
- **Objective: Jointly model the distribution of R (the CDRs) conditioned on the structure of the antibody-antigen complex (C).**

  * Assume the CDR to be generated contains m amino acids, indexed from l+1 to l+m:

$$R = \{(s_j, x_j, O_j) \mid j = l+1, \ldots, l+m\}.$$

$$C = \{(s_i, x_i, O_i) \mid i \in \{1, \ldots, N\} \setminus \{l+1, \ldots, l+m\}\}.$$

Methods

II: Diffusion Process

Forward Diffusion Process:

- The forward diffusion process is **a training mechanism**.
- Gradually **adds noise to the data** (e.g., amino acid sequences, positions, and orientations) to perturb it, effectively simulating a distribution of noisy states.
- The goal is **to create a sequence of intermediate noisy states from the original data**, eventually reaching a near-uniform prior distribution.

Generative Diffusion Process:

- The generative diffusion process is **a sampling mechanism**.
- **Starts from the noisy state** (generated by the prior distribution), **iteratively transforms the prior to match the desired distribution.**
- The goal is **to iteratively denoise a noisy input**, starting from a random prior distribution, **to generate a realistic and functional antibody sequence** and structure conditioned on the given antigen and antibody framework.

  * Let the intermediate state of amino acid j at time step t be denoted as: $\{s_j^t, x_j^t, O_j^t\}$
  * The sequence and structure sampled at step t are represented as:

$$R^t = \{s_j^t, x_j^t, O_j^t \mid j = l+1, \ldots, l+m\}$$

Where: s_j^t represents the amino acid type, x_j^t represents the coordinate, and O_j^t represents the orientation.

Ho, J., Jain, A., & Abbeel, P. (2020b, June 19). *Denoising diffusion probabilistic models*. arXiv.org. https://arxiv.org/abs/2006.11239

Methods

II: Diffusion Process - Forward Diffusion Process

- **Stepwise Forward Diffusion:**

  At each timestep t, **noise is added** to the data x_{t-1} to generate the noisy data x_t

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(x_t \mid \sqrt{1-\beta_t} \cdot x_{t-1}, \beta_t \mathbf{I}\right)$$

Where β_t: Variance schedule controlling the amount of noise added at timestep t.
$\mathbf{I}$: Identity matrix, representing isotropic Gaussian noise.

- **Direct Forward Diffusion from Initial Data:**

  The noisy data x_t can also be **computed directly from the original data** x_0

$$q(x_t \mid x_0) = \mathcal{N}\left(x_t \mid \sqrt{\bar{\alpha}_t} \cdot x_0, (1-\bar{\alpha}_t)\mathbf{I}\right)$$

Where $\bar{\alpha}_t = \prod^{t}(1-\beta_i)$ represents the cumulative noise schedule up to timestep t.

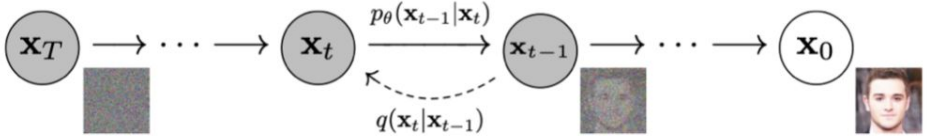


Figure 2: The directed graphical model considered in this work.

- **A Gaussian (Normal) Distribution:**

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

with mean μ and standard deviation σ

  - The Probability Density Function (PDF) is:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- **A multinomial Distribution (Probability mass function):**

$$P(X = [x_1, x_2, \ldots, x_k]) = \frac{n!}{x_1! x_2! \cdots x_k!} \cdot p_1^{x_1} \cdot p_2^{x_2} \cdots p_k^{x_k}$$

Where $\mathbf{X} = [x_1, x_2, \ldots, x_k]$ is the outcome of the multinomial distribution (counts in k categories), $n = \sum_{i=1}^{k} x_i$ gives the total number of trials, and p_i gives the probability of category i, with $\sum_{i=1}^{k} p_i = 1$ and $\frac{n!}{x_1! x_2! \cdots x_k!}$ is multinomial coefficient accounting for the different ways outcomes can be distributed among categories.

Methods

II: Diffusion Process - Generative Diffusion Process

- **Conditional Probability in the Generative Model**

$$p_\theta(x_{t-1}|x_t)$$

This represents the probability of x_{t-1} reconstructing (less noisy data) given x_t (a noisier version of the data).

- **Reverse Diffusion Process**

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1}|x_t)$$

Where

$p(x_T)$: The prior distribution, typically a standard Gaussian $\mathcal{N}(0, I)$.

$p_\theta(x_{t-1}|x_t)$: The learned reverse conditional probabilities.

- **Role of the Neural Network**
  - The neural network $\theta$is trained to minimize the difference between the predicted reverse transition $p_\theta(x_{t-1}|x_t)$ and the true reverse transition derived from the forward process $q(x_{t-1}|x_t, x_0)$
  - During training, it optimizes the evidence lower bound (ELBO), often simplified to:

$$\mathcal{L} = \mathbb{E}_t \left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2 \right]$$

Where ϵ: The true noise added in forward diffusion.

$\epsilon_\theta(x_t, t)$: The noise predicted by the model.

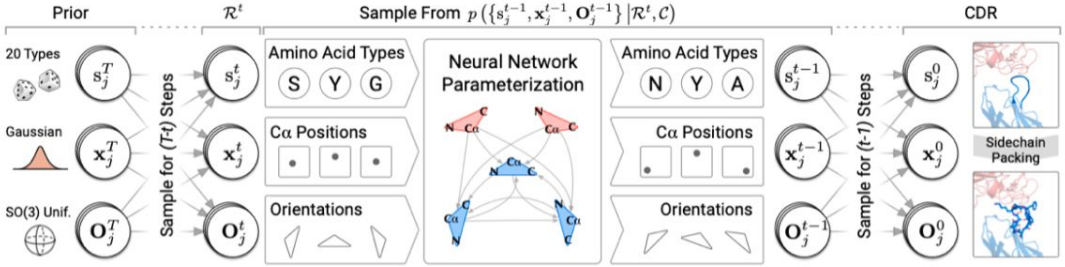


Figure 3: Illustration of the generative diffusion process. At each step, the network takes the current CDR state as input and parameterizes the distribution of the CDR's sequences, positions, and orientations for the next step. In the end, full-atom structures are constructed by the side-chain packing algorithm.

$$\{s_j^t, x_j^t, O_j^t\}$$

- **Key Parameters:**
  - **Amino acid types**
  - **A-coordinates**
  - **Amino acid orientations**

Methods

II: Diffusion Process

Multinomial Diffusion for Amino Acid Types:

- **The forward diffusion process**

$$q(s_j^t|s_j^{t-1}) = \text{Multinomial}\left((1-\beta_t^{\text{type}})\cdot \text{onehot}(s_j^{t-1}) + \beta_t^{\text{type}}\cdot\frac{1}{20}\cdot\mathbf{1}\right)$$

Where $\text{onehot}(s_j^{t-1})$ converts the amino acid type into a 20-dimensional one-hot vector. $\mathbf{1}$ represents all one vectors, and β_t^{type} gives the probability of resampling another amino acid uniformly from 20 types. As $t \to T, \beta_t^{\text{type}}$ approaches $\mathbf{1}$, leading the distribution closer to a uniform distribution.

- **For the training:**

The probability density to perturb s_j^0 at timestep t is:

$$q(s_j^t|s_j^0) = \text{Multinomial}\left(\alpha_t^{\text{type}}\cdot \text{onehot}(s_j^0) + (1-\alpha_t^{\text{type}})\cdot\frac{1}{20}\cdot\mathbf{1}\right) \quad \text{Eqs. (*)}$$

Where $\alpha_t^{\text{type}} = \prod_{\tau=1}^{t}(1-\beta_\tau^{\text{type}})$

Methods

II: Diffusion Process

Multinomial Diffusion for Amino Acid Types:

- **The generative diffusion process**

$$p(s_j^{t-1}|R^t, C) = \text{Multinomial}\left(F(R^t, C)[j]\right)$$

Where $F(\cdot)[j]$ is a neural network that predicts the amino acid probabilities based on 1) structure context C(antigen and antibody framework); 2) previous CDR state R^t (sequence, positions, and orientations).

- **For the training:**

The generative diffusion process aims to approximate the posterior $q(s_j^{t-1}|s_j^t, s_j^0)$ by minimizing the expected Kullback-Leibler (KL) divergence:

$$L_{\text{type}}^t = \mathbb{E}_{R^t \sim p}\left[\frac{1}{m}\sum_j D_{\text{KL}}\left(q(s_j^{t-1} \mid s_j^t, s_j^0) \,\middle\|\, p(s_j^{t-1} \mid R^t, C)\right)\right]. \quad *$$

Where $q(s_j^{t-1}|s_j^t, s_j^0)$ is the posterior derived from the forward process. $p(s_j^{t-1}|R^t, C)$ predicts distribution from the generative process.

Methods

II: Diffusion Process

Diffusion for C_α Coordinate:

To standardize the distribution of atom coordinates, the C_α coordinates, represented as vectors $\mathbf{x}_j \in \mathbb{R}^3$, are scaled and shifted such that they approximately follow a standard normal distribution.

- **The forward diffusion for the normalized C_α coordinate $\mathbf{x}_j$**

  Stepwise diffusion:

$$q(\mathbf{x}_j^t \mid \mathbf{x}_j^{t-1}) = \mathcal{N}\left(\mathbf{x}_j^t \mid \sqrt{1-\beta_{\text{pos}}^t} \cdot \mathbf{x}_j^{t-1}, \beta_{\text{pos}}^t \mathbf{I}\right)$$

  Cumulative distribution from the initial state:

$$q(\mathbf{x}_j^t \mid \mathbf{x}_j^0) = \mathcal{N}\left(\mathbf{x}_j^t \mid \sqrt{\bar{\alpha}_{\text{pos}}^0} \cdot \mathbf{x}_j^0, (1-\bar{\alpha}_{\text{pos}}^0)\mathbf{I}\right) \quad \text{Eqs. (**)}$$

Where β_{pos}^t controls the rate of diffusion and increases from 0 to 1, $\mathbf{I}$ is the Identity matrix, representing isotropic Gaussian noise.

$$\bar{\alpha}_t^{\text{pos}} = \prod_{\tau=1}^{t} (1-\beta_\tau^{\text{pos}})$$

Methods

II: Diffusion Process

Diffusion for C_α Coordinate:

- **The generative diffusion process reverses the forward process, and is defined as:**

$$p(\mathbf{x}_j^{t-1} \mid R^t, C) = \mathcal{N}\left(\mathbf{x}_j^{t-1} \mid \boldsymbol{\mu}_p(R^t, C), \beta_{\text{pos}}^t \mathbf{I}\right)$$

Where the mean $\boldsymbol{\mu}_p(R^t, C)$ is calculated as:

$$\boldsymbol{\mu}_p(R^t, C) = \frac{1}{\sqrt{\bar{\alpha}_{\text{pos}}^t}} \left(\mathbf{x}_j^t - \frac{\beta_{\text{pos}}^t}{\sqrt{1 - \bar{\alpha}_{\text{pos}}^t}} G(R^t, C)[j] \right).$$

Where $G(\cdot)[j]$ is a neural network that predicts the Gaussian noise $\boldsymbol{\epsilon}_j \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, added during the forward process, and we have:

$$\mathbf{x}_j^t = \sqrt{\bar{\alpha}_t^{\text{pos}}} \cdot \mathbf{x}_j^0 + \sqrt{1 - \bar{\alpha}_{\text{pos}}^0} \cdot \boldsymbol{\epsilon}_j.$$

- **The training objective:**

  The generative diffusion process is trained to minimize the expected mean squared error (MSE) between the predicted noise $G(R^t, C)[j]$ and the true noise $\boldsymbol{\epsilon}_j$

$$L_{\text{pos}}^t = \mathbb{E}\left[\frac{1}{m} \sum_j \left\| \boldsymbol{\epsilon}_j - G(R^t, C) \right\|^2 \right].$$

Methods

II: Diffusion Process

SO(3) Denoising for Amino Acid Orientations:

The amino acid orientations are represented as elements on the special orthogonal group SO(3).

- **The perturbed orientation distribution after t steps is defined as:**

$$q(\mathbf{O}_j^t \mid \mathbf{O}_j^0) = \mathrm{IG}_{\mathrm{SO}(3)}\left(\mathbf{O}_j^t \mid \mathrm{ScaleRot}(\sqrt{\bar{\alpha}_{\mathrm{ori}}^t}, \mathbf{O}_j^0), 1 - \bar{\alpha}_{\mathrm{ori}}^t\right) \quad \text{Eqs. (***)}$$

Where $\mathrm{IG}_{\mathrm{SO}(3)}$ is the Isotropic Gaussian distribution on SO(3), parameterized by a mean rotation and scalar variance. ScaleRot scales the rotation matrix by modifying the rotation angle while keeping the rotation axis fixed. $\bar{\alpha}_{\mathrm{ori}}^t = \prod_{\tau=1}^{t}(1 - \beta_{\mathrm{ori}}^{\tau})$ controls the variance and increases with step t.

- **The conditional distribution for the generative process is given by:**

$$p(\mathbf{O}_j^{t-1} \mid R^t, C) = \mathrm{IG}_{\mathrm{SO}(3)}\left(\mathbf{O}_j^{t-1} \mid H(R^t, C)[j], \beta_{\mathrm{ori}}^t\right)$$

Where $H(R^t, C)[j]$ is a neural network that predicts the denoised orientation matrix for amino acid j.

Methods

II: Diffusion Process

SO(3) Denoising for Amino Acid Orientations:

- **Training Objective for Orientations:**

  The goal is to align the predicted orientation $\hat{\mathbf{O}}_j^{t-1}$ with the real orientation $\mathbf{O}_j^0$ by minimizing the expected discrepancy, measured using the Frobenius norm:

$$L_{\text{ori}}^t = \mathbb{E}\left[\frac{1}{m}\sum_j \left\|(\mathbf{O}_j^0)^\top \hat{\mathbf{O}}_j^{t-1} - \mathbf{I}\right\|_F^2\right]$$

Where $\hat{\mathbf{O}}_j^{t-1} = H(R^t, C)[j]$ is the predicted orientation for amnio acid j.

Methods

II: Diffusion Process

The Overall Training Objective:

The overall training objective combines the loss functions for amino acid types L^t_{type}, coordinates L^t_{pos}, and orientations L^t_{ori}:

$$L = \mathbb{E}_{t\sim\text{Uniform}(1,\ldots,T)}\left[L^t_{\text{type}} + L^t_{\text{pos}} + L^t_{\text{ori}}\right].$$

The Training Procedure:

1. Sample a time step $t \sim \text{Uniform}(1, \ldots, T)$
2. Add noise to the training sample to generate noisy states $\{s^t_j, \mathbf{x}^t_j, \mathbf{O}^t_j\}_{j=l+1}^{l+m}$ using the diffusion processes defined in Eqs. (*), (**), and (***)
3. Compute the total loss L using the noisy data.
4. Backpropagate the loss to update model parameters.

Methods

III: Parameterization with Neural Networks

- **Purpose:** The neural network architectures are designed to **encode the CDR state at a given time step t** along with the context structure:

$$\{s_j^t, \mathbf{x}_j^t, \mathbf{O}_j^t\}_{j=l+1}^{l+m} \cup \{s_i, \mathbf{x}_i, \mathbf{O}_i\}_{i \in \{1,\ldots,N\} \setminus \{l+1,\ldots,l+m\}}$$

and subsequently denoise the CDR **amino acid types (F), positions (G), and orientations (H).**

- **Steps:**
  1. **Embedding Generation:**
     - Single embeddings: amino acid properties;
     - Pairwise embeddings: distance, dihedrals.
  2. **Feature Transformation:**
     - IPA Network for orientation-aware encoding.
  3. **Denoising:**
     - Amnio acid types: posterior probabilities.
     - Coordinates: local to global frame projection.
     - Orientation: SO(3) matrix for updating
  4. **Equivariance Properties:** Networks invariant to rotation and translation

Methods

IV:Sampling Algorithms

- **Steps:**
  1. **Initialization:**
     - $s_j^T \sim \text{Uniform}(20), \quad \mathbf{x}_j^T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_3), \quad \mathbf{O}_j^T \sim \text{Uniform}(\text{SO}(3)).$
     - Normalize coordinate for C_α positions.
  2. **Iterative Sampling:**
     - Denoise amino acid types, coordinates, and orientations using the generative diffusion kernel.
     - Continue until t = 0
  3. **3D Structure Construction**:
     - Build backbone atoms: N, C_α , C, O, and C_β
     - Use Rosetta for side-chain packing.
     - Refine structure with AMBER99 force field in OpenMM.
- **Applications:**
  - **Fix-Backbone Sequence Design:** Fix structure, sample sequences.
  - **Antibody Optimization:** Add noise and refine existing antibodies.

Experiments

Experiments

I: Sequence-Structure Co-design

- **Dataset:** SAbDab database, test set includes 19 complexes (e.g., SARS-CoV-2, MERS)
- **Evaluation:** Removed original CDRs and sampled sequences/structures, and Compared DiffAb to RAbD (100 samples per CDR).
- **Metrics:** *
  - IMP (Improvement in Binding Energy Percentage): % design with better binding energy
  - RMSD (Root-Mean-Square Deviation): Structural deviation
  - AAR: sequence recovery rate
- **Key results:**
  - **Higher AAR** (Amino Acid Recovery Rate): better sequence recovery across all CDRs
  - **Greater Diversity:** higher RMSD in CDR-H3
  - **Competitive IMP**: comparable binding energy without Rosetta-based training.

Table 1: Evaluation of the generated antibody CDRs (sequence-structure co-design) by RAbD and our DiffAb model.

CDR	Method	AAR	RMSD	IMP	CDR	Method	AAR	RMSD	IMP
H1	RAbD	22.85%	2.261Å	43.88%	L1	RAbD	34.27%	1.204Å	46.81%
	DiffAb	65.75%	1.188Å	53.63%		DiffAb	56.67%	1.388Å	45.58%
H2	RAbD	25.50%	1.641Å	53.50%	L2	RAbD	26.30%	1.767Å	56.94%
	DiffAb	49.31%	1.076Å	29.84%		DiffAb	59.32%	1.373Å	49.95%
H3	RAbD	22.14%	2.900Å	23.25%	L3	RAbD	20.73%	1.624Å	55.63%
	DiffAb	26.78%	3.597Å	23.63%		DiffAb	46.47%	1.627Å	47.32%

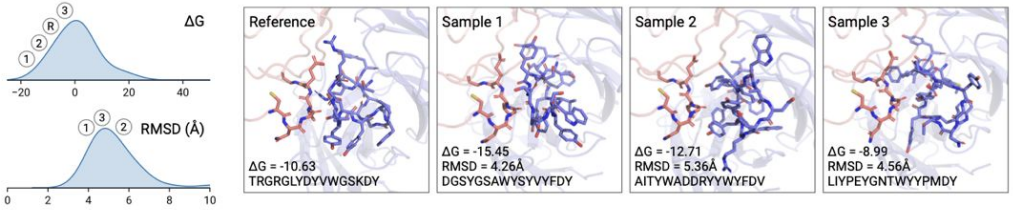


Figure 4: Examples of CDR-H3 designed by the sequence-structure co-design method and the distribution of their interaction energy and RMSD. The antigen-antibody template is derived from PDB:7chf, where the antigen is SARS-CoV-2 RBD. Sample 1 has better complementarity to the antigen while Sample 3 fits the antigen worse. This could explain their difference in the binding energy (ΔG).

Experiments

II: Fix-Backbone Sequence Design and Structure Prediction

- **Task:**
  - Design CDR sequences with given backbone structures (a constrained sampling problem).
- **Evaluation:**
  - Baseline: FixBB (Rosetta-based sequence design software).
  - Metric: AAR (Amino Acid Recovery Rate).
- **Key results:**
  - **Better AAR**: DiffAb outperforms FixBB for all CDRs (Table 2)
  - **Shows strong modeling of conditional sequence probabilities.**
  - High AAR attributed to training clusters, but non-H3 CDRs contribute less to specificity.

Table 2: Comparison of FixBB and DiffAb in terms of amino acid recovery (AAR) in the fix-backbone CDR design task. DiffAb achieves higher AAR. The AAR of DiffAb on CDR-H3 is lower than other CDRs since H3 is much more versatile.

CDR	Method	AAR	CDR	Method	AAR
H1	FixBB	37.14%	L1	FixBB	33.80%
	DiffAb	87.83%		DiffAb	86.63%
H2	FixBB	43.08%	L2	FixBB	28.54%
	DiffAb	79.70%		DiffAb	88.91%
H3	FixBB	30.74%	L3	FixBB	17.92%
	DiffAb	59.48%		DiffAb	78.69%

Table 3: The accuracy of CDR structures predicted by DiffAb in RMSD.

CDR	RMSD
H1	0.901Å
H2	1.044Å
H3	3.246Å
L1	1.365Å
L2	1.321Å
L3	1.492Å

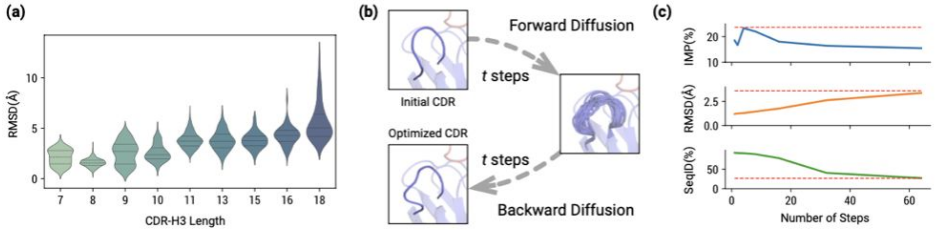


Figure 5: **(a)** RMSD of predicted CDR-H3 structures grouped by lengths. **(b)** The antibody optimization algorithm first perturbs the initial CDR for t steps using the forward diffusion process and then denoises it by the backward diffusion process into the optimized CDR. **(c)** IMP, RMSD, and SeqID of the CDRs optimized with different numbers of steps. Dashed lines represent the results of *de novo* design. When $t = 4$, the optimized CDRs reach an IMP score close to *de novo* CDRs but remain structurally similar to the original one.

Experiments

III: Antibody Optimization

Antibody Optimization

- **Task:**
  - Optimized CDR-H3 regions using generative diffusion by perturbing sequences for t steps and denoising.
- **Evaluation metrics:**
  - IMP: Improvement in binding energy (max: 23.29% at t = 4)
  - RMSD: Structural deviation
  - SeqID: Sequence similarity to original.
- **Key results:**
  - **Achieved improved binding energy while maintaining sequence similarity,** critical for pharmaceutical applications.

Table 4: Evaluation of optimized CDR-H3s with different numbers of optimization steps. In contrast to redesigning the CDR, the optimization method can improve binding energy while *keeping the optimized CDR similar to the original one*. Figure 5c shows the line plot of the results.

t	IMP	RMSD	SeqID
1	18.52%	1.194Å	92.42%
2	16.67%	1.252Å	91.61%
4	23.29%	1.290Å	91.16%
8	22.01%	1.447Å	88.78%
16	18.02%	1.759Å	78.43%
32	16.43%	2.623Å	40.58%
64	15.47%	3.380Å	27.30%
T	23.63%	3.597Å	26.78%

Experiments

III: Antibody Optimization

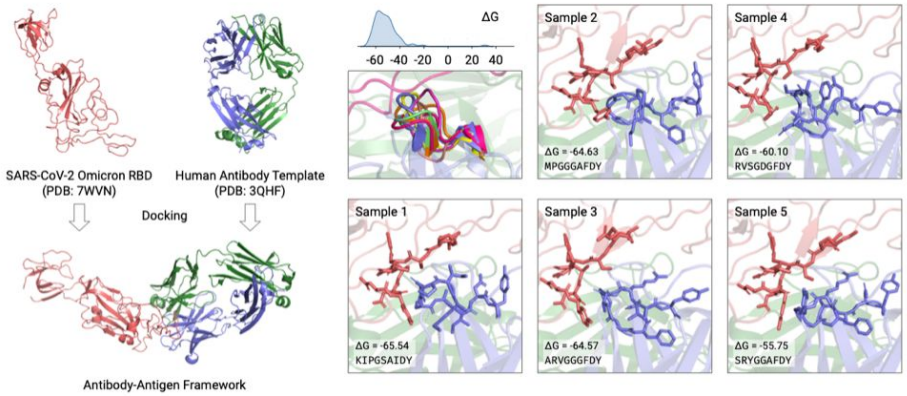


Figure 6: A human antibody framework docked to SARS-CoV-2 Omicron RBD using HDOCK. CDR-H3s are designed based on the docking structure.

Antibody Docking and Binding Energy

- **Setup:**
  - Docking of SARS-CoV-2 Omicron RBD (PDB: 7VWN) with a human antibody template (PDB: 3QHF) using HDock.
  - Designed CDR-H3 sequences optimized based on the docking structure.
- **Results:**
  - **Top 5 optimized antibodies have strong binding energies (ΔG), ranging from -55.75 to -65.54 kcal/mol.**
  - Example optimized sequences:
    - Sample 1: KIPGSAIDY (ΔG = -65.54)

- **Visualization:**
  - Superimposed structures show alignment between designed antibodies and the antigen-binding regions.
  - Binding energies (ΔG) validate the high - quality design.
- **Key Take-away:**
  - Designed antibodies demonstrate strong binding to the SARS-CoV-2 Omicron RBD with high binding affinities, showcasing the potential of the generative diffusion approach in designing therapeutically effective antibodies

Experiments

IV: Design Without Bound Antibody Frameworks

- **Objective:**
  - To tackle the challenging task of designing antibodies without a known binding pose against the antigen, a scenario where no bound antibody framework is available.
- **Methodology:**
  - **Antibody Template Preparation:**
    - Started with an existing antibody structure and removed the CDR-H3 region (as it is the most variable and specific to binding).
    - Retained the other conserved CDRs from the antibody framework.
- **Docking**:
  - Used **HDOCK** docking software to **dock the modified antibody template to the target antigen** (SARS-CoV-2 Omicron RBD, PDB: 7VWN) and **generate an antibody-antigen complex.**

- **Antibody Design**:
  - **Once docked, the problem was reduced to the earlier framework (antibody optimization).**
  - Designed new CDR-H3 sequences and structures, and re-designed other CDRs using the generative diffusion model.
- **Validation:**
  - Evaluated the generated antibodies using the binding energy distribution.
  - Demonstrated that the designed antibodies have reasonable binding energies, although their effectiveness could not be confirmed without a reference antibody.
- **Key results:**
  - The approach is generalizable to antibodies without pre-existing binding poses, showing potential for applications in novel antibody design against unbound antigens.

Conclusions & Recommendations

Conclusions & Recommendations:

- **Conclusions**:
  - Developed a diffusion-based generative model for antibody design.
  - Supports various antibody design tasks with competitive performance.

- **Limitations**:
  - **Dependency on Bound Structures**:
    1. Requires an antibody framework bound to the target antigen.
    2. Future work: Extend to unbound antibody design.
  - **Biological Validation**:
    1. Unclear if generated antibodies can be produced in the lab and bind effectively.
    2. Further efforts needed for biological efficacy.

- **Recommendations:**
  - **Validation with Experimental Data**: include experimental validation of the designed antibodies, such as in vitro binding affinity assays or structural characterization, to strengthen the claims about their therapeutic potential.
  - **Applications Beyond CDR-H3**: Explore the potential to design or optimize other CDRs and regions, especially for antibodies where non-H3 regions contribute significantly to binding.

Appendix

Appendix:

I: Kullback-Leibler (KL) divergence

The loss function that measures the divergence between the true and learned reverse processes. This KL-based loss ensures that the learned generative process approximates the true reverse process, enabling effective denoising during inference.

1.1 Continuous KL Divergence

The KL divergence between two continuous probability distributions P and Q over a variable x is defined as:

$$D_{\mathrm{KL}}(P\|Q) = \int P(x) \log \frac{P(x)}{Q(x)}\, dx$$

where:

- $P(x)$: The true distribution.
- $Q(x)$: The approximate distribution.

The integral accounts for the entire domain of x. A smaller KL divergence indicates a closer match between $P(x)$ and $Q(x)$.

1.2 Discrete KL Divergence

For discrete probability distributions, the KL divergence is given by:

$$D_{\mathrm{KL}}(P\|Q) = \sum_{x} P(x) \log \frac{P(x)}{Q(x)}$$

where the summation is taken over all possible values of x. This formulation is commonly used in scenarios where events or categories are discrete, such as in classification tasks or natural language processing.

1.3 KL Divergence in Diffusion Models

In diffusion models, the KL divergence plays a critical role in comparing the forward diffusion process q with the learned reverse process p_θ. The loss function is defined as:

$$\mathcal{L}_{\mathrm{KL}} = \mathbb{E}_{q(x_t|x_0)} \left[D_{\mathrm{KL}} \left(q(x_{t-1}|x_t, x_0) \| p_\theta(x_{t-1}|x_t) \right) \right]$$

where:

- $q(x_t|x_0)$: The forward diffusion process, describing how noise is added to the data.
- $q(x_{t-1}|x_t, x_0)$: The true reverse process conditioned on forward noise.
- $p_\theta(x_{t-1}|x_t)$: The learned reverse generative process, which aims to denoise the data.

Appendix:

II: Metric Calculations

2.1 IMP (Improvement in Binding Energy Percentage)

IMP measures the percentage of designed CDRs that have better binding energy (ΔG) compared to the original CDRs. It is calculated as:

$$\text{IMP} = \frac{\text{Number of CDRs with Improved Binding Energy}}{\text{Total Number of Designed CDRs}} \times 100$$

where:

- **Improved Binding Energy:** A lower ΔG value compared to the original CDR indicates better binding affinity.
- **Total Number of Designed CDRs:** The total count of antibody CDRs designed during the process.

2.2 RMSD (Root-Mean-Square Deviation)

RMSD quantifies the average structural deviation between the predicted and reference structures of antibody CDRs. It is expressed as:

$$\text{RMSD} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \|x_i^{\text{predicted}} - x_i^{\text{reference}}\|^2}$$

where:

- $x_i^{\text{predicted}}$: The predicted position of atom i.
- $x_i^{\text{reference}}$: The reference (true) position of atom i.
- N: The total number of atoms being compared.

A lower RMSD value indicates higher structural similarity between the predicted and reference CDR structures.

2.3 AAR (Amino Acid Recovery Rate)

AAR measures the percentage of amino acids in the designed CDR sequence that match those in the reference CDR sequence. It is calculated as:

$$\text{AAR} = \frac{\text{Number of Matched Amino Acids}}{\text{Total Number of Amino Acids in CDR}} \times 100$$

where:

- **Matched Amino Acids:** The amino acids in the designed CDR that are identical to those in the reference CDR.
- **Total Number of Amino Acids in CDR:** The length of the CDR sequence.

A higher AAR indicates better recovery of the reference sequence in the designed CDR.

Appendix:

II: Metric Calculations - Energy Calculations:

3.1 Definition of Binding Energy

The binding energy is defined as:

$$\Delta G = G_{\text{bound}} - (G_{\text{antibody}} + G_{\text{antigen}})$$

where:

- G_{bound}: Free energy of the antibody-antigen complex.
- G_{antibody}: Free energy of the antibody in its unbound state.
- G_{antigen}: Free energy of the antigen in its unbound state.

A lower ΔG value indicates a stronger binding interaction between the antibody and antigen.

3.2 Methods for Calculating Binding Energy

Binding energy can be calculated using various computational approaches:

3.2.1 Rosetta Energy Functions

Rosetta provides energy functions to calculate binding energy using tools such as 'InterfaceAnalyzer'. The steps are:

1. Dock the antibody and antigen to form a complex.
2. Calculate the total energy of the complex (G_{bound}).
3. Separate the antibody and antigen and calculate their individual energies (G_{antibody} and G_{antigen}).
4. Compute the binding energy:

$$\Delta G = G_{\text{bound}} - (G_{\text{antibody}} + G_{\text{antigen}})$$

3.3 Simplified Binding Energy Formula

In many computational tools, the binding energy is approximated as:

$$\Delta G = E_{\text{interaction}} - E_{\text{unbound}}$$

where:

- $E_{\text{interaction}}$: Interaction energy between the antibody and antigen in the bound state.
- E_{unbound}: Sum of energies when the molecules are separated.